Reproduce-then-Extend — Civil-War Onset (Fearon & Laitin 2003) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Fearon, J. D. & Laitin, D. D. (2003). "Ethnicity, Insurgency, and Civil War." *American Political Science Review* 97(1):75–90. A **logistic regression** of civil-war onset on 11 country-year predictors (Sambanis 2006 imputed data; Harvard Dataverse doi: 10.7910/DVN/KRKWK8). Onset is a **rare event** (~1.6%); this example illustrates **out-of-sample classification** and the right evaluation metric for imbalanced data.

1 Background

Fearon and Laitin ask **what conditions make civil war likely** worldwide from 1945 to 1999. Their central aim is to adjudicate between two explanations: civil war as a product of ethnic and religious **grievance**, versus civil war as a product of conditions that favor **insurgency** – weak states, poverty, rough terrain, and populations governments cannot control. They use **logistic regression** of civil-war onset on structural covariates to test which set of factors predicts onset. The **primary conclusion** is that insurgency-favoring conditions (low GDP per capita, large population, mountainous terrain, political instability, new states) strongly predict civil war, while ethnic and religious fractionalization do **not** – a direct challenge to the ethnic-grievance thesis.

2 Data and codebook

Unit of analysis: country-year; n = 7,140 (116 civil-war onsets, ~1.6%).

Variable	Definition
`warstds`	civil-war onset: 1 in the first year of a new civil war **(outcome)**
`warhist`	prior civil-war history in the country
`ln_gdpen`	log GDP per capita (lagged one year)
`lpopns`	log national population
`lmtnest`	log % of the country that is mountainous terrain
`ncontig`	1 if the state has noncontiguous territory
`oil`	1 if fuel exports exceed one-third of total exports (oil exporter)
`nwstate`	1 in the first two years of a newly independent state
`inst3`	political instability: a >=3-point Polity change in the prior 3 years
`pol4`	Polity score (-10 autocracy to +10 democracy)
`ef`	ethnolinguistic fractionalization
`relfrac`	religious fractionalization

3 Descriptive analysis

Before modeling we profile the data end to end: provenance and variable roles, the outcome's rarity, the shape and missingness of every predictor, each predictor's marginal association with onset, and the collinearity structure of the design matrix. The through-line is that this is a **rare-event binary** problem, and the descriptives tell us which evaluation metrics and modeling choices that fact forces on us.

```
d <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day1_",
  "intro_and_evaluation/02_fearon_laitin/data/Sambnis",
  "Imp.csv"))
fl <- c("warhist", "ln_gdpen", "lpopns", "lmtnest", "ncontig", "oil", "nwstate",
        "inst3", "pol4", "ef", "relfrac")
d <- d[complete.cases(d[, c("warstds", fl)]), c("warstds", fl)]
```

3.1 Dataset and provenance

The data are the Sambanis (2006) replication of Fearon & Laitin (2003), distributed as a **multiply-imputed** country-year panel on the Harvard Dataverse (doi:10.7910/DVN/KRKWK8). The **unit of analysis is a country-year** spanning 1945–1999; after restricting to the outcome and the eleven Table-1 covariates we retain 7140 complete country-years. The outcome `warstds` is the **onset** of a new civil war (coded 1 only in the war's first year), so the sample is dominated by peace-years and onsets are rare by construction.

```
cat(sprintf("rows (country-years): %d\n", nrow(d)))
```

```
## rows (country-years): 7140
```

```
cat(sprintf("columns: 1 outcome + %d predictors\n", length(fl)))
```

```
## columns: 1 outcome + 11 predictors
```

```
cat(sprintf("civil-war onsets: %d  (base rate %.2f%%)\n",
            sum(d$warstds), 100 * mean(d$warstds)))
```

```
## civil-war onsets: 116  (base rate 1.62%)
```

A subtlety worth flagging up front: three nominally binary indicators (`ncontig`, `nwstate`, `inst3`) were **imputed**, so although they range on $[0, 1]$ they take fractional values for the ~9–11% of rows that were filled in — we treat them as continuous $[0, 1]$ scores below, not clean dummies.

```
roles <- data.frame(
  variable = c("warstds","warhist","ln_gdpen","lpopns","lmtnest","ncontig",
               "oil","nwstate","inst3","pol4","ef","relfrac"),
  role = c("outcome", rep("predictor", 11)),
  type = c("binary","binary","continuous","continuous","continuous",
           "imputed [0,1]","binary","imputed [0,1]","imputed [0,1]",
           "continuous","continuous","continuous"),
  description = c(
    "civil-war onset (1 = first war-year)","prior civil-war history",
    "log GDP per capita (lagged)","log population","log % mountainous terrain",
    "noncontiguous territory","oil exporter (fuel > 1/3 exports)",
    "new state (first two years)","political instability (>=3 Polity change)",
    "Polity score (-10..+10)","ethnolinguistic fractionalization",
    "religious fractionalization"))
knitr::kable(roles, caption = "Variable dictionary: outcome and predictors.")
```

Table 1: Variable dictionary: outcome and predictors.

variable	role	type	description
warstds	outcome	binary	civil-war onset (1 = first war-year)
warhist	predictor	binary	prior civil-war history
ln_gdpen	predictor	continuous	log GDP per capita (lagged)
lpopns	predictor	continuous	log population
lmtnest	predictor	continuous	log % mountainous terrain
ncontig	predictor	imputed [0,1]	noncontiguous territory
oil	predictor	binary	oil exporter (fuel > 1/3 exports)
nwstate	predictor	imputed [0,1]	new state (first two years)
inst3	predictor	imputed [0,1]	political instability (>=3 Polity change)
pol4	predictor	continuous	Polity score (-10..+10)
ef	predictor	continuous	ethnolinguistic fractionalization
relfrac	predictor	continuous	religious fractionalization

3.2 The outcome in depth

`warstds` is binary, so its "distribution" is a **base rate**: onsets occur in only 1.62% of country-years. The stats table below reports it as a 0/1 variable; for a Bernoulli variable the mean *is* the base rate and the (positive) skew and heavy kurtosis simply restate how lopsided the classes are.

```
sstat <- function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), mean = mean(x), sd = sd(x), min = min(x),
    Q1 = q[[1]], median = q[[2]], Q3 = q[[3]], max = max(x),
    IQR = q[[3]] - q[[1]], skew = e1071::skewness(x),
    kurt = e1071::kurtosis(x))
}
round(t(sapply(d["warstds"], sstat)), 3)
```

```
##             n  mean    sd min Q1 median Q3 max IQR  skew   kurt
## warstds 7140 0.016 0.126   0  0      0  0   1   0 7.651 56.552
```

```
par(mar = c(4, 4.5, 2.5, 1))
bp <- barplot(table(d$warstds), names.arg = c("no onset", "onset"),
              col = c("grey75", "#1f78b4"), border = "white",
              main = "Outcome: warstds (base rate 1.6%)", ylab = "country-years")
text(bp, table(d$warstds), labels = table(d$warstds), pos = 3, xpd = NA, cex = .9)
```

Implication for evaluation. With a 1.6% base rate, **accuracy is meaningless** — the "always predict peace" baseline already scores 98.4%. Class imbalance is exactly why the prediction section below reports the **ROC** and, more honestly, the **precision-recall** curve, whose no-skill baseline is the base rate itself. No transform applies to a binary outcome; the modeling response to rarity is metric choice (and, in extensions, resampling or penalties), not a re-expression of y.

3.3 Univariate predictor distributions

We summarize all eleven predictors in one table (rounding so nothing overflows the margin), then show the numeric ones as faceted histograms.

```
psum <- function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), n_miss = sum(is.na(x)), mean = mean(x), sd = sd(x),
    min = min(x), Q1 = q[[1]], median = q[[2]], Q3 = q[[3]],
```

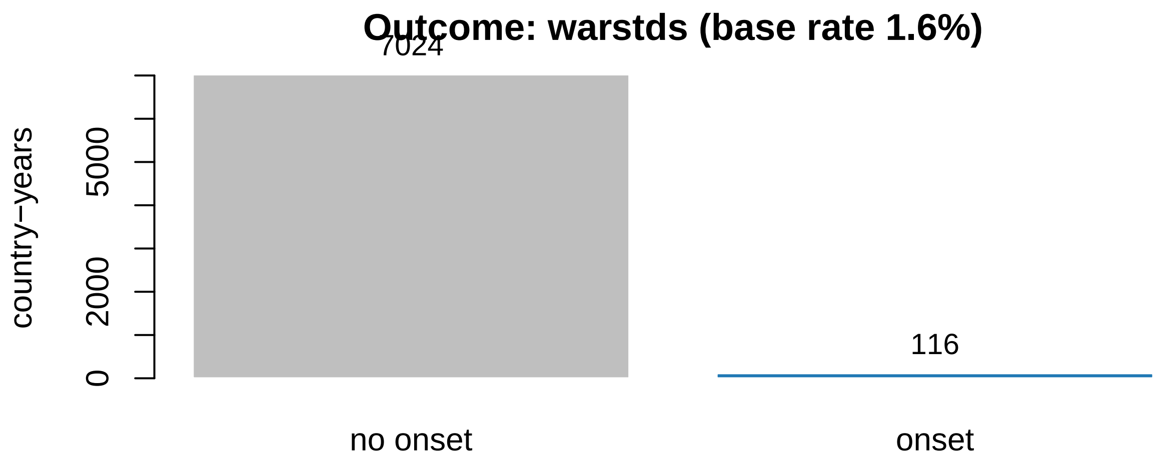

Figure 1: Class balance of the rare onset outcome. Peace-years outnumber onsets roughly 60-to-1, so a model that always predicts 'no war' is 98.4% accurate yet useless.

```
    max = max(x), skew = e1071::skewness(x))
}
round(t(sapply(d[fl], psum)), 3)
```

```
##               n n_miss   mean    sd     min     Q1 median     Q3    max   skew
## warhist  7140      0  0.171 0.377   0.000  0.000  0.000  0.000  1.000  1.747
## ln_gdpen 7140      0  0.761 0.995  -3.037  0.032  0.845  1.393  4.067  0.088
## lpopns   7140      0 15.746 1.625  11.014 14.817 15.701 16.742 20.947  0.082
## lmtnest  7140      0  2.183 1.339   0.000  1.194  2.282  3.215  4.557 -0.255
## ncontig  7140      0  0.175 0.364   0.000  0.000  0.000  0.155  1.000  1.789
## oil      7140      0  0.118 0.323   0.000  0.000  0.000  0.000  1.000  2.363
## nwstate  7140      0  0.028 0.159   0.000  0.000  0.000  0.000  1.000  5.934
## inst3    7140      0  0.154 0.343   0.000  0.000  0.000  0.118  1.000  2.015
## pol4     7140      0 -0.097 7.264 -10.000 -7.000 -1.000  8.000 10.000  0.210
## ef       7140      0  0.464 0.252   0.004  0.238  0.463  0.664  1.000  0.007
## relfrac  7140      0  0.364 0.209   0.000  0.186  0.352  0.540  0.783  0.183
```

```
library(ggplot2); library(tidyr)
long <- pivot_longer(d[fl], everything(), names_to = "var",
                     values_to = "value")
ggplot(long, aes(value)) +
  geom_histogram(bins = 30, fill = "#1f78b4", colour = "white",
                 linewidth = .15) +
  facet_wrap(~ var, scales = "free", ncol = 4) +
  labs(x = NULL, y = "count") +
  theme_bw(base_size = 8)
```

The truly binary predictors `warhist` (prior war) and `oil` (oil exporter) are best read as frequency tables:

```
bin_tab <- function(v) {
  tb <- table(factor(d[[v]], levels = c(0, 1)))
  data.frame(variable = v, n_0 = tb[[1]], n_1 = tb[[2]],
             pct_1 = round(100 * tb[[2]] / sum(tb), 1))
}
knitr::kable(do.call(rbind, lapply(c("warhist", "oil"), bin_tab)),
             row.names = FALSE,
```

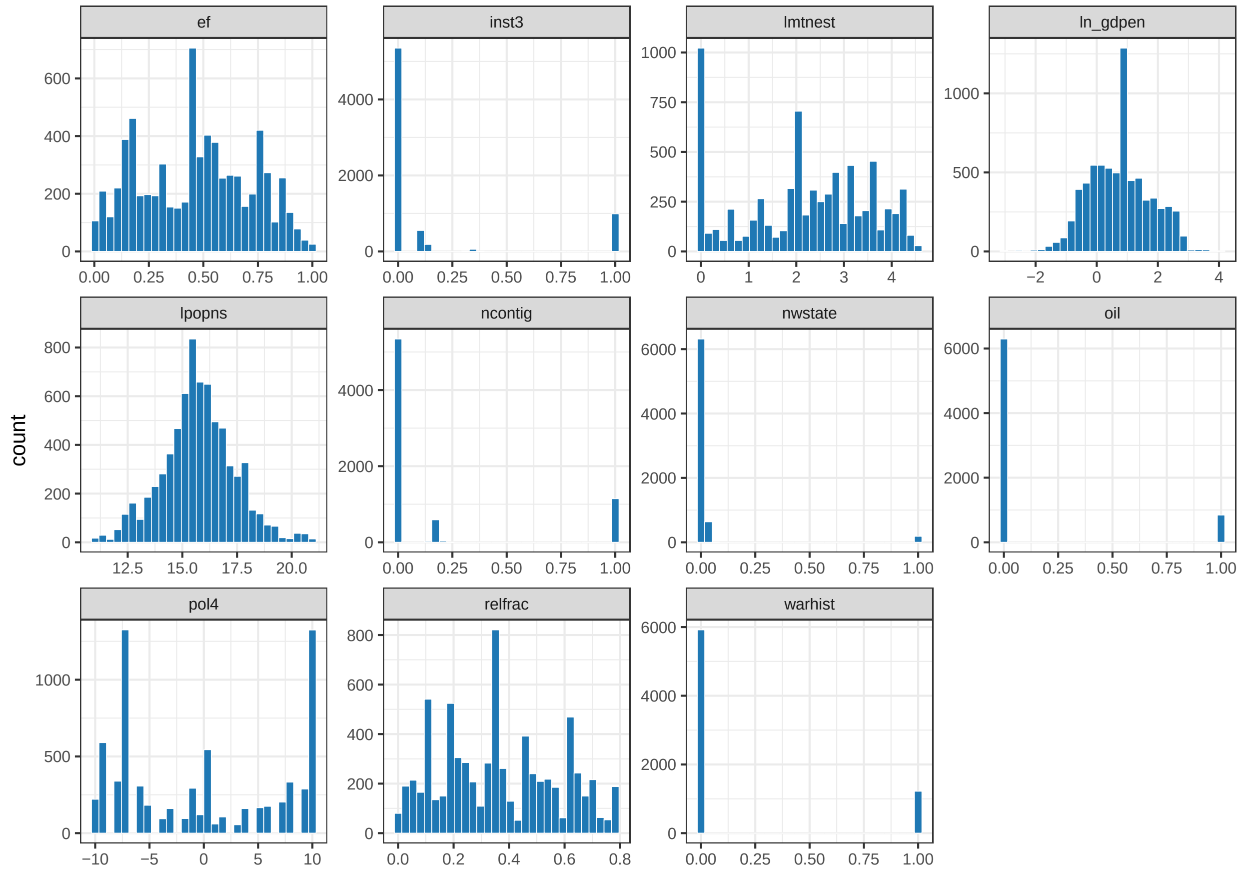

Figure 2: Univariate distributions of the eleven predictors. Logged economic/terrain variables are roughly symmetric; the imputed indicators (ncontig, nwstate, inst3) and oil/warhist pile up at 0.

```
            caption = "Frequency of the two clean binary predictors.")
```

Table 2: Frequency of the two clean binary predictors.

variable	n_0	n_1	pct_1
warhist	5919	1221	17.1
oil	6295	845	11.8

Read-out. The logged variables (`ln_gdpen`, `lpopns`, `lmtnest`) are close to symmetric — logging has already tamed the raw right-skew of income, population and terrain, which is why the published model uses them on the log scale. `pol4` (Polity) is bimodal, clustering at the autocratic and democratic poles. `ef`/`relfrac` are bounded fractionalization indices on $[0, 1]$. About a fifth of country-years carry a prior-war history and roughly a quarter are oil exporters.

3.4 Missing-data analysis

Because the file is a completed multiple imputation, there are essentially **no missing cells** in the modeling columns — the imputation is precisely how Sambanis handled the extensive missingness in the raw Fearon–Laitin covariates.

```
miss <- data.frame(
  variable = c("warstds", fl),
  n_missing = sapply(d[c("warstds", fl)], function(x) sum(is.na(x))),
  pct_missing = round(100 * sapply(d[c("warstds", fl)],
                    function(x) mean(is.na(x))), 2))
cat(sprintf("complete cases: %d of %d (%.1f%%)\n",
            sum(complete.cases(d)), nrow(d),
            100 * mean(complete.cases(d))))
```

```
## complete cases: 7140 of 7140 (100.0%)
```

```
knitr::kable(miss, row.names = FALSE,
             caption = "Missingness by variable (post-imputation).")
```

Table 3: Missingness by variable (post-imputation).

variable	n_missing	pct_missing
warstds	0	0
warhist	0	0
ln_gdpen	0	0
lpopns	0	0
lmtnest	0	0
ncontig	0	0
oil	0	0
nwstate	0	0
inst3	0	0
pol4	0	0
ef	0	0
relfrac	0	0

The reproduction takes complete cases (a no-op here), so the imputation's uncertainty is not propagated — a known simplification worth naming: the standard errors below treat the imputed values as observed.

3.5 Bivariate relationships with the outcome

For a binary outcome the natural bivariate summary is **how each predictor differs between onset and non-onset years**. We report the group means, a standardized mean difference (Cohen's d), and a two-sample t-test p-value.

```
biv <- t(sapply(fl, function(v) {
  x1 <- d[[v]][d$warstds == 1]; x0 <- d[[v]][d$warstds == 0]
  sp <- sqrt(((length(x1) - 1) * var(x1) +
              (length(x0) - 1) * var(x0)) /
             (length(x1) + length(x0) - 2))
  c(mean_onset = mean(x1), mean_peace = mean(x0),
    cohen_d = (mean(x1) - mean(x0)) / sp,
    p_value = t.test(x1, x0)$p.value)
}))
knitr::kable(round(biv, 3),
             caption = "Marginal association of each predictor with onset.")
```

Table 4: Marginal association of each predictor with onset.

	mean_onset	mean_peace	cohen_d	p_value
warhist	0.190	0.171	0.050	0.608
ln_gdpen	0.167	0.771	-0.608	0.000
lpopns	16.333	15.736	0.367	0.000
lmtnest	2.641	2.176	0.348	0.000
ncontig	0.197	0.175	0.061	0.548
oil	0.138	0.118	0.062	0.540
nwstate	0.122	0.027	0.601	0.002
inst3	0.419	0.150	0.788	0.000
pol4	-1.996	-0.065	-0.266	0.001
ef	0.546	0.463	0.331	0.001
relfrac	0.405	0.363	0.200	0.034

```
smd <- biv[, "cohen_d"]; smd <- smd[order(smd)]
par(mar = c(4, 6, 2.5, 1))
barplot(smd, horiz = TRUE, las = 1, cex.names = 0.8,
        col = ifelse(smd > 0, "#1f78b4", "#e31a1c"),
        main = "Onset vs non-onset", xlab = "std. mean difference (Cohen's d)")
abline(v = 0, col = "grey50")
```

The largest separations are exactly the **insurgency-favoring** conditions: onset years have **lower GDP per capita**, **larger populations**, **more mountainous terrain**, more **prior-war history** and more **instability**. Ethnolinguistic and religious fractionalization (`ef`, `relfrac`) barely move — the descriptive foreshadowing of Fearon & Laitin's headline null on ethnic grievance.

3.6 Multivariate structure and collinearity

Finally we check whether the predictors are entangled with one another, since collinearity inflates standard errors and destabilizes the logit.

```
cm <- cor(d[fl])
cml <- as.data.frame(as.table(cm))
names(cml) <- c("v1", "v2", "r")
cml$v1 <- factor(cml$v1, levels = fl)
```

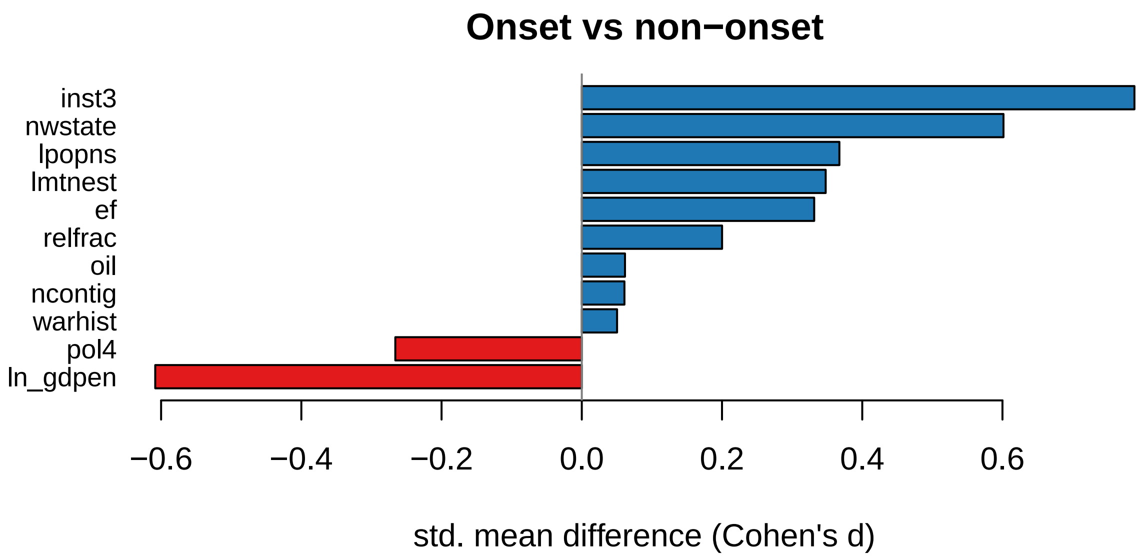

Figure 3: Standardized mean difference (onset minus non-onset, in SD units) for each predictor. Onset years are poorer, more populous, more mountainous, and carry more prior-war history and instability.

```
cml$v2 <- factor(cml$v2, levels = rev(fl))
ggplot(cml, aes(v1, v2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.2) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white",
                       high = "#1f78b4", midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
# variance inflation factors from the reproduction design (no car dependency)
X <- d[fl]
vif <- sapply(fl, function(v) {
  r2 <- summary(lm(reformulate(setdiff(fl, v), v), data = X))$r.squared
  1 / (1 - r2)
})
mx <- max(abs(cor(d[fl])[upper.tri(cor(d[fl]))]))
cat(sprintf("largest absolute pairwise correlation: %.2f\n", mx))
```

```
## largest absolute pairwise correlation: 0.46
```

```
round(sort(vif, decreasing = TRUE), 2)
```

```
## ln_gdpen     pol4       ef  ncontig   lpopns  relfrac      oil  lmtnest  warhist
##     1.67     1.50     1.41     1.30     1.24     1.22     1.21     1.13     1.09
##    inst3  nwstate
##     1.05     1.02
```

Every pairwise $|r|$ is below the 0.7 flag and all **VIFs sit near 1** (well under the customary 5), so multicollinearity is not a concern: the eleven covariates carry largely independent information and the logit coefficients below are individually identifiable.

Descriptive verdict. The picture is a **highly imbalanced binary outcome** whose onset years separate cleanly on insurgency conditions but not on ethnic grievance, over a clean, low-collinearity, fully-imputed

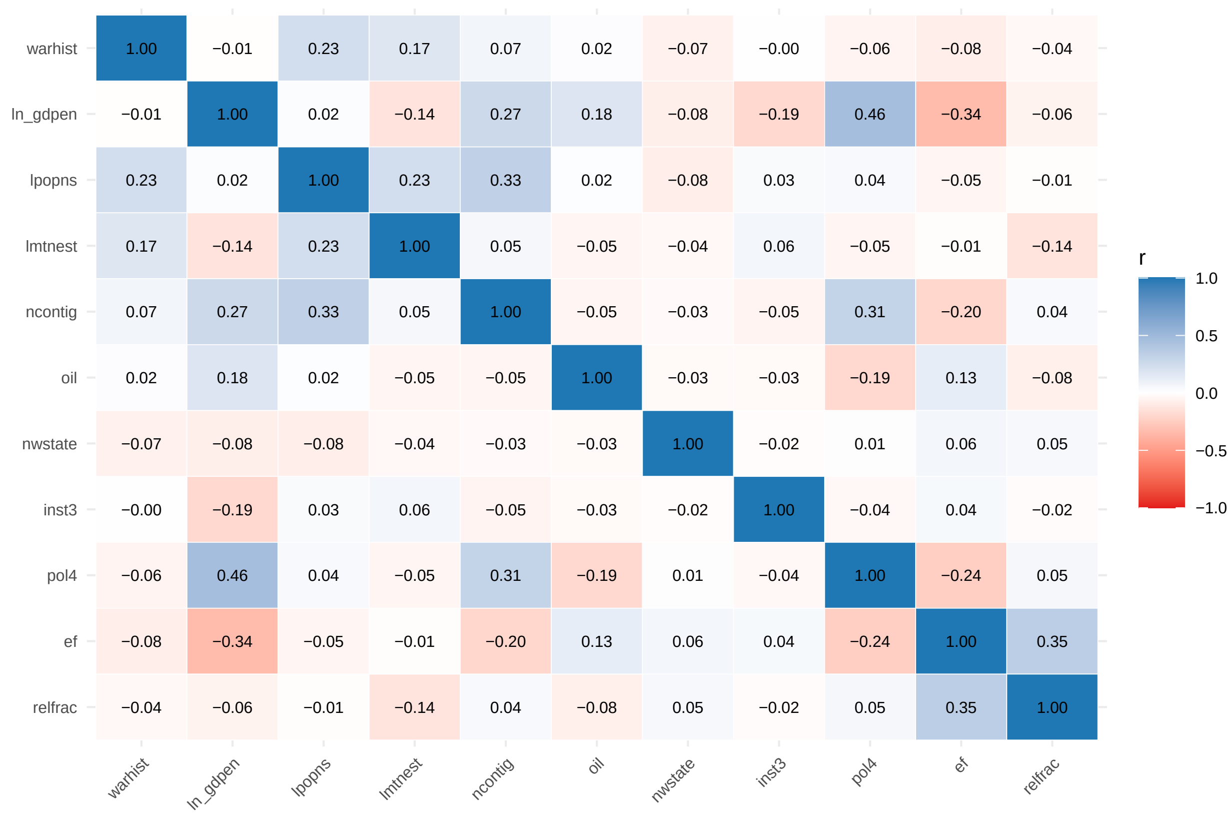

Figure 4: Pearson correlations among the eleven predictors. No |r| exceeds 0.7, so the design matrix is well conditioned.

design. That combination is what motivates the choices ahead: fit the published logit for the explanatory claim, then evaluate its *predictions* out of sample with **precision-recall**, not accuracy.

4 Reproduce the published regression

```
fl_logit <- glm(warstds ~ warhist + ln_gdpen + lpopns + lmtnest + ncontig + oil +
                  nwstate + inst3 + pol4 + ef + relfrac, data = d, family = binomial)
summary(fl_logit)
```

```
##
## Call:
## glm(formula = warstds ~ warhist + ln_gdpen + lpopns + lmtnest +
##     ncontig + oil + nwstate + inst3 + pol4 + ef + relfrac, family = binomial,
##     data = d)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.46582    1.11539  -7.590 3.20e-14 ***
## warhist      0.02247    0.25379   0.089  0.92946
## ln_gdpen    -0.35196    0.11791  -2.985  0.00284 **
## lpopns       0.18788    0.06815   2.757  0.00584 **
## lmtnest      0.18706    0.08049   2.324  0.02013 *
## ncontig      0.26477    0.28232   0.938  0.34831
## oil          0.33689    0.30155   1.117  0.26391
## nwstate      1.86032    0.32037   5.807 6.37e-09 ***
## inst3        1.36212    0.20709   6.577 4.79e-11 ***
## pol4        -0.02435    0.01781  -1.367  0.17170
## ef           0.44421    0.42340   1.049  0.29411
## relfrac      0.59413    0.50782   1.170  0.24202
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1185.9  on 7139  degrees of freedom
## Residual deviance: 1057.2  on 7128  degrees of freedom
## AIC: 1081.2
##
## Number of Fisher Scoring iterations: 7
```

Confirmation against the published study. The coefficients carry the signs and significance reported in Fearon & Laitin's Table 1: civil war is more likely in **poorer** countries (`ln_gdpen` negative), **more populous** ones (`lpopns` positive), **more mountainous** terrain (`lmtnest` positive), **new states** (`nwstate` positive) and amid **political instability** (`inst3` positive), while ethnic/religious fractionalization are not significant — their central claim that *insurgency conditions, not ethnic grievance, drive civil war.*

5 Out-of-sample prediction with the regression

We fit on a random 70% and score the held-out 30%. With a rare outcome, accuracy is useless (predicting "no war" everywhere scores 98%); the honest evaluation is the **ROC** and especially the **precision-recall** curve.

```
set.seed(2026)
train <- sample(nrow(d), floor(0.7 * nrow(d)))
test  <- setdiff(seq_len(nrow(d)), train)
```

```r
fit  <- glm(warstds ~ ., data = d[train, ], family = binomial)
pred <- predict(fit, d[test, ], type = "response")   # predicted onset probability
y    <- d$warstds[test]

roc <- roc.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve =
↪  TRUE)
pr  <- pr.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve = TRUE)
cat(sprintf("Held-out test: %d country-years, onset base rate %.1f%%\n", length(test),
↪  100 * mean(y)))
```

```
## Held-out test: 2142 country-years, onset base rate 1.6%
```

```r
cat(sprintf("  Out-of-sample ROC-AUC = %.3f\n  Out-of-sample PR-AUC  = %.3f  (a random
↪  model scores %.3f)\n",
            roc$auc, pr$auc.integral, mean(y)))
```

```
##   Out-of-sample ROC-AUC = 0.819
##   Out-of-sample PR-AUC  = 0.068  (a random model scores 0.016)
```

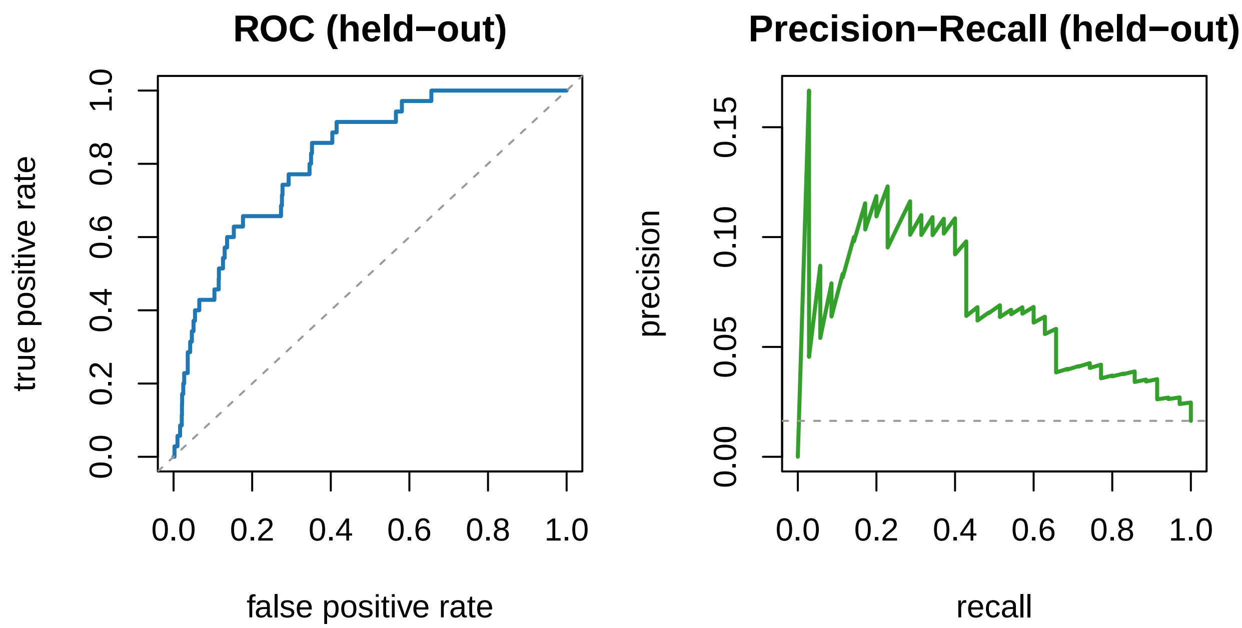

Figure 5: Out-of-sample ROC curve (left) and precision-recall curve (right) for the held-out logit predictions.

The ROC-AUC looks strong (~0.8), but the **PR-AUC** is far more modest — the honest picture when only 1.6% of cases are onsets. This is the key evaluation lesson: *on imbalanced data, ROC-AUC flatters a model and precision-recall tells the truth.*

6 Takeaway

This example illustrates **out-of-sample classification with a published regression** and the **evaluation-metric choice** that matters most for rare events. The Fearon–Laitin logit predicts onset above chance out of sample, but reporting precision–recall (not accuracy or ROC alone) is what keeps the assessment honest. (A later example — Gibler–Braithwaite — extends this rare-event setting with tree-based learners.)

7 Recommended exercises

1. Replace the single split with stratified 5-fold CV and report mean AUC-PR (the outcome is ~1.6% positive, so use AUC-PR, not accuracy).
2. Fit an L1-penalized logistic regression and compare its AUC-PR and retained variables to the full logit.
3. Plot the calibration (reliability) curve of the predicted probabilities and comment on over/under-confidence.